Homework 08 Solutions

Prerith Arunkumar (prerith2)

Margin geometry for a fixed linear score in two and four dimensions, and a peer review of a Homework 7 LDA/QDA analysis skill.

Use of AI tools. I used Claude (Claude Code) to draft the `R` code and exposition for Questions 1 and 2 and to run them. Every number below comes from the code output shown in this report, and I checked the numbers quoted in the text against it. For Question 3, I ran the assigned skill with an AI agent as the assignment requires; the peer review itself is written by me without AI assistance.

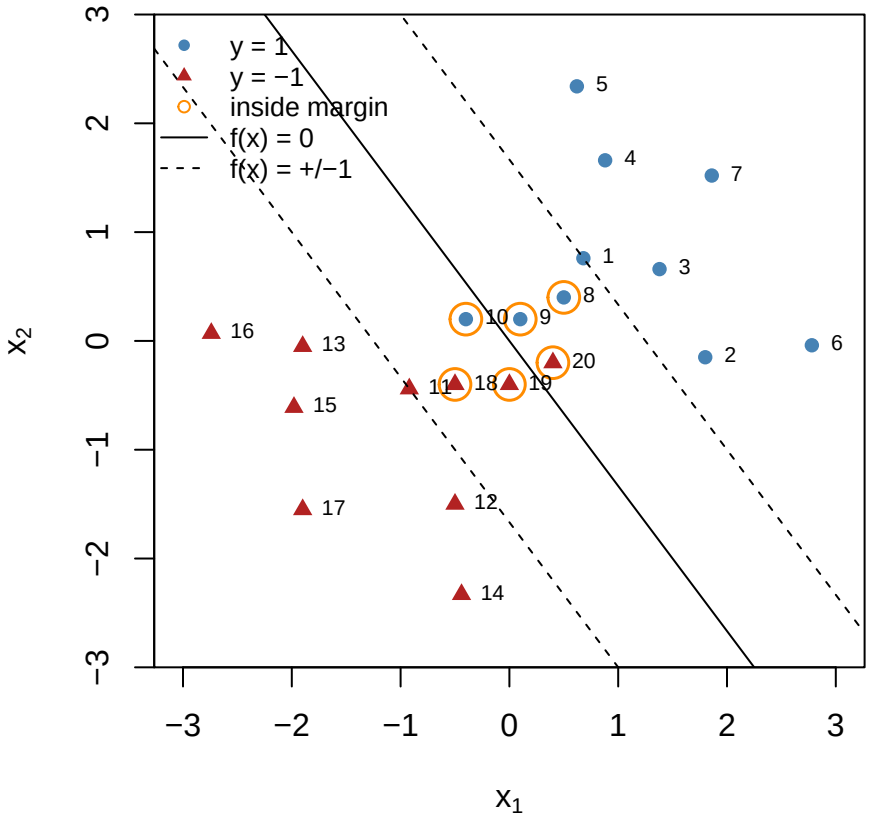
Question 1: Boundary and margin

```
d <- read.csv("data/margin-geometry-20.csv")
w <- c(0.8, 0.6)                                  # ||w|| = 1
d$f <- drop(as.matrix(d[, c("x1", "x2")]) %*% w)
d$distance <- abs(d$f) / sqrt(sum(w^2))           # distance to f(x) = 0
d$yf <- d$y * d$f                                 # functional margin
tol <- 1e-8
d$position <- ifelse(abs(d$yf - 1) < tol, "on",
                     ifelse(d$yf < 1, "inside", "beyond"))
d$misclassified <- d$yf < 0
knitr::kable(d[, c("observation", "x1", "x2", "y", "distance", "yf", "position", "misclassified")],
             digits = 2, row.names = FALSE)
```

observation	x1	x2	y	distance	yf	position	misclassified
1	0.68	0.76	1	1.00	1.00	on	FALSE
2	1.80	-0.15	1	1.35	1.35	beyond	FALSE
3	1.38	0.66	1	1.50	1.50	beyond	FALSE
4	0.88	1.66	1	1.70	1.70	beyond	FALSE
5	0.62	2.34	1	1.90	1.90	beyond	FALSE
6	2.78	-0.04	1	2.20	2.20	beyond	FALSE
7	1.86	1.52	1	2.40	2.40	beyond	FALSE
8	0.50	0.40	1	0.64	0.64	inside	FALSE
9	0.10	0.20	1	0.20	0.20	inside	FALSE
10	-0.40	0.20	1	0.20	-0.20	inside	TRUE
11	-0.92	-0.44	-1	1.00	1.00	on	FALSE
12	-0.50	-1.50	-1	1.30	1.30	beyond	FALSE
13	-1.90	-0.05	-1	1.55	1.55	beyond	FALSE
14	-0.44	-2.33	-1	1.75	1.75	beyond	FALSE
15	-1.98	-0.61	-1	1.95	1.95	beyond	FALSE
16	-2.74	0.07	-1	2.15	2.15	beyond	FALSE
17	-1.90	-1.55	-1	2.45	2.45	beyond	FALSE
18	-0.50	-0.40	-1	0.64	0.64	inside	FALSE
19	0.00	-0.40	-1	0.24	0.24	inside	FALSE
20	0.40	-0.20	-1	0.20	-0.20	inside	TRUE

Because $\|(0.8, 0.6)\|_2 = 1$, the distance to the boundary is $|f(x)|$. An observation is beyond its margin when $yf(x) > 1$, on it when $yf(x) = 1$, and inside it when $yf(x) < 1$.

```
cols <- ifelse(d$y == 1, "steelblue", "firebrick")
par(mar = c(4, 4, 1, 1), xaxs = "i", yaxs = "i")
plot(d$x1, d$x2, pch = ifelse(d$y == 1, 16, 17), col = cols, asp = 1,
     xlim = c(-3.2, 3.2), ylim = c(-3, 3), xlab = expression(x[1]), ylab = expression(x[2]))
for (k in c(-1, 0, 1))                              # 0.8 x1 + 0.6 x2 = k
  abline(a = k / 0.6, b = -0.8 / 0.6, lty = ifelse(k == 0, 1, 2))
ins <- d$position == "inside"
points(d$x1[ins], d$x2[ins], cex = 2.2, col = "darkorange", lwd = 1.5)
text(d$x1, d$x2, d$observation, pos = 4, cex = 0.7)
legend("topleft", c("y = 1", "y = -1", "inside margin", "f(x) = 0", "f(x) = +/-1"),
       pch = c(16, 17, 1, NA, NA), lty = c(NA, NA, NA, 1, 2),
       col = c("steelblue", "firebrick", "darkorange", "black", "black"), bty = "n", cex = 0.8)
```




```
table(factor(d$position, levels = c("inside", "on", "beyond")))
```

```
inside     on beyond
     6      2     12
```

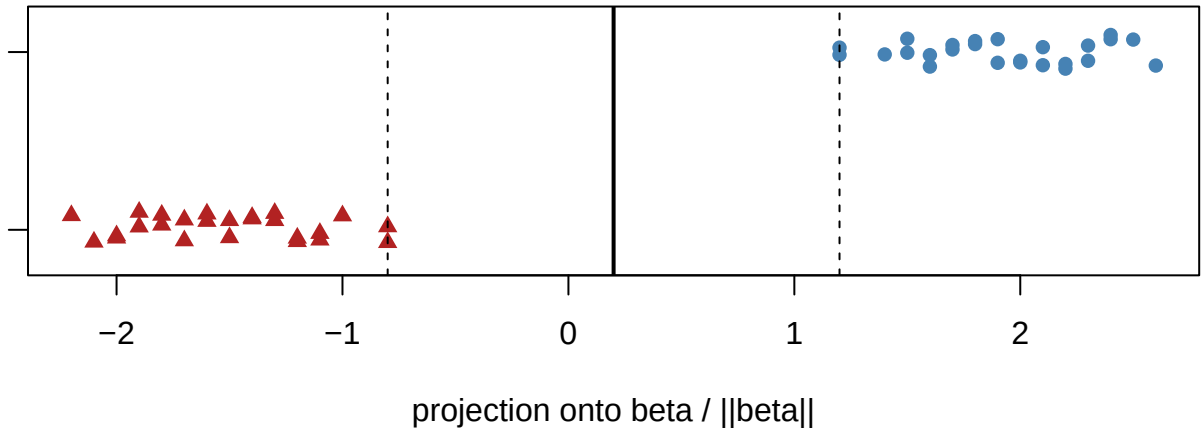
```
d$observation[d$misclassified]
```

```
[1] 10 20
```

Six observations (8, 9, 10, 18, 19, 20) are inside the margin, two (1 and 11) lie exactly on it, and the other twelve are beyond it. Observations 10 and 20 have $yf(x) < 0$, so they are on the wrong side of the boundary and are misclassified; the rest of the observations inside the margin are correctly classified margin violations.

Question 2: Margin in four dimensions

```
q <- read.csv("data/margin-geometry-4d.csv")
X <- as.matrix(q[, c("x1", "x2", "x3", "x4")])
beta0 <- -1; beta <- c(1, 2, -2, 4)
norm_b <- sqrt(sum(beta^2))
proj <- drop(X %*% beta) / norm_b                  # coordinate along beta / ||beta||
boundary <- -beta0 / norm_b                         # f(x) = 0 on this axis
signed_dist <- proj - boundary                      # = f(x) / ||beta||
par(mar = c(4, 1, 1, 1))
stripchart(list(`y = -1` = proj[q$y == -1], `y = 1` = proj[q$y == 1]), method = "jitter",
           pch = c(17, 16), col = c("firebrick", "steelblue"), xlab = "projection onto beta / ||beta||")
abline(v = boundary, lwd = 2)
abline(v = boundary + c(-1, 1) * min(q$y * signed_dist), lty = 2)
```




```
c(norm_beta = norm_b, boundary = boundary,
  M = min(q$y * signed_dist), all_correct = all(q$y * signed_dist > 0))
```

```
  norm_beta    boundary           M all_correct
        5.0         0.2         1.0         1.0
```

The solid line is the boundary at projection $-\beta_0/\|\beta\| = 0.2$. All observations are correctly classified, and the closest ones are a distance $M = 1$ from the boundary (dashed lines).

```
fm <- q$y * (beta0 + drop(X %*% beta))
c(min_functional_margin = min(fm))
```

```
min_functional_margin
                    5
```

```
c(beta0 = beta0, beta) / min(fm)
```

```
beta0
 -0.2   0.2   0.4  -0.4   0.8
```

The smallest functional margin under the given coefficients is $5 = \|\beta\|M$. Dividing by 5 gives the same boundary with $\min_i y_i f(\mathbf{x}_i) = 1$: $\beta = (0.2, 0.4, -0.4, 0.8)^\top$ and $\beta_0 = -0.2$. Then $\|\beta\| = 1$, consistent with $M = 1/\|\beta\| = 1$.

Question 3: Peer review of a Homework 7 skill

Assigned repository: mazy2005/stat432-fall2026 (Homework 7 report).

TODO (Prerith): skill run. Describe the run: the prompt given to the agent, time and token use, whether a steer message was sent or the run was stopped, and what the skill produced.

TODO (Prerith): peer review, written without AI. Brief overall assessment, then three major items, each with what you observed, why it matters, and what you recommend.

GitHub Issue: TODO (link to the Issue opened in mazy2005/stat432-fall2026).